Computational Research Scientist, Simmons Cancer Center

UT Southwestern Medical Center · Dallas, TX, US

Job ID (this archive)	HI-029
Employer	UT Southwestern Medical Center
Location(s)	Dallas, TX, US
Career stage (archive tag)	Research / graduate degree
Location group	DFW area
Majors (archive tags)	Biomedical Engineering, Electrical Engineering, Computer Science, Data Science
Pay listed	Not listed
Employment type	Full-time
Employer requisition	965040
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://jobs.utsouthwestern.edu/job/23689439/computational-research-scientist-simmons-cancer-center-dallas-tx/
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Course sheet refs	HI-029

Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

Job description

WHY UT SOUTHWESTERN?

With over 75 years of excellence in Dallas-Fort Worth, Texas, UT Southwestern is committed to excellence, innovation, teamwork, and compassion. As a world-renowned medical and research center, we strive to provide the best possible care, resources, and benefits for our valued employees. Ranked as the number 1 hospital in Dallas-Fort Worth according to U.S. News & World Report , we invest in you with opportunities for career growth and development to align with your future goals. Our highly competitive benefits package offers healthcare, PTO and paid holidays, on-site childcare, wage, merit increases and so much more. We invite you to be a part of the UT Southwestern team where you'll discover a culture of teamwork, professionalism, and a rewarding career!

JOB SUMMARY

Leads the design and implementation of High-Performance Scientific Computing infrastructure. Performs and supports scientific research using technical knowledge of software, software development, complex databases, high-performance computing, and/or hardware in a complex computing environment. Performs scientific research using knowledge of formalisms and algorithms from mathematics, statistics, and/or computer science.

The Center for Cellular Therapies and Cancer Immunology in the Harold C. Simmons Comprehensive Cancer Center is looking to hire a Computational Research Scientist at the interface of machine learning, cancer genomics, and immunology. You will build and apply computational methods that turn large-scale multi-omic and functional datasets into validated therapeutic hypotheses, while working side by side with experimental and translational scientists to bring those hypotheses to the clinic.

This is a senior, high-autonomy position suited to someone who wants their models to drive biological discovery to therapies. You will help set scientific direction, mentor junior computational and experimental scientists, and serve as the analytical backbone of a program developing novel immunotherapies within a new center.

What you will do

- Build predictive models that integrate genomic, transcriptomic, epigenomic, single-cell, and functional-screen data to nominate novel therapeutic targets in cancer and the tumor immune microenvironment.

- Drive immunotherapy engineering by applying machine learning to antigen and neoantigen discovery, de novo protein design, and the optimization of engineered immune cells.
- Develop methods for interpreting high-dimensional and perturbation data, including deep learning approaches to sequence, structure, and single-cell readouts.
- Partner closely with wet-lab scientists to design experiments, prioritize candidates for validation, and iterate rapidly between computation and the bench.
- Lead and mentor trainees and staff, set analytical standards, and help shape the scientific roadmap of the program.
- Communicate results through high-impact publications, presentations, and collaborations across UT Southwestern's research and clinical community.

Areas of focus

Depending on your interests and strengths, your work can center on identifying novel cancer immunotherapy targets and/or developing novel therapeutics. Approaches may include:

- Machine learning applied to multi-omic, single-cell, and functional data
- Clinical-molecular correlations that link genotype and tumor biology to therapeutic response
- De novo protein design
- Computational proteomics

Education

o Ph.D. in Bioinformatics, Computational Biology, Biomedical Engineering, or related field

o Master's degree in computer science, Electrical Engineering, or related field

Experience

o 2 years experience in driving computational biology/machine learning projects.

o 4 years experience in driving computational biology/machine learning projects with Master's Degree.

o Post-doctoral fellowship preferred.

BENEFITS

UT Southwestern is proud to offer a competitive and comprehensive benefits package to eligible employees. Our benefits are designed to support your overall wellbeing, and include:

PPO medical plan, available day one at no cost for full-time employee-only coverage

100% coverage for preventive healthcare-no copay

Paid Time Off, available day one

Retirement Programs through the Teacher Retirement System of Texas (TRS)

Paid Parental Leave Benefit

Wellness programs

Tuition Reimbursement

Public Service Loan Forgiveness (PSLF) Qualified Employer

Learn more about these and other UTSW employee benefits!

EXPERIENCE AND EDUCATION

Required

Education

PhD In Computer Science, Electrical Engineering, or related field or

Master's Degree In Computer Science, Electrical Engineering, or related field

Experience

2 years Experience in software development/engineering and/or HPC systems management with PhD. or

4 years Experience in software development/engineering and/or HPC systems management with Master's Degree.

JOB DUTIES

Designs and implements the high-performance scientific computing infrastructure for for large, multi-institution projects, including software applications, workflows, servers, databases, long-term and short-term storage solutions, and cloud services.

Serves as a domain expert for research and industry collaborations on biomedical computational methods, algorithms and platforms. Subject matter expert in multiple areas of computing technology.

Leads and develops proof-of-concepts (PoCs) for solutions applied to biomedical and clinical science use-cases by working closely with industry collaborators.

Designs, develops, implements and optimizes algorithms and scientific workflows for high dimensional data in biomedical and clinical sciences.

Develops software and methods to explore, analyze and visualize biological data sets.

Demonstrates and craft applications of cutting-edge visual computing technologies and connect them to core infrastructure.

Report research results in journal articles, conference papers, and technical manuals.

Mentor and train staff, postdoctoral fellows, and graduate students.

Supports faculty and students in adapting computational strategies to leverage high-performance computing, working with a range of systems and technologies such as compute cluster, parallel file systems, high speed interconnects, GPU-based computing and data servers.

Performs other duties as needed

SECURITY AND EEO STATEMENT

Security

This position is security-sensitive and subject to Texas Education Code 51.215, which authorizes UT Southwestern to obtain criminal history record information. To the extent this position requires the holder to research, work on, or have access to critical infrastructure as defined in Section 117.001(2) of the Texas Business and Commerce Code, the ability to maintain the security or integrity of the critical infrastructure is a minimum qualification to be hired and to continue to be employed in the position.

EEO

UT Southwestern Medical Center is committed to an educational and working environment that provides equal opportunity to all members of the University community. As an equal opportunity employer, UT Southwestern prohibits unlawful discrimination, including discrimination on the basis of race, color, religion, national origin, sex, sexual orientation, gender identity, gender expression, age, disability, genetic information, citizenship status, or veteran status.

Notes for ECS 1100 students

How this connects to your major

Data Science: healthcare data analysis, data engineering, statistics, AI or computational biology; Computer Science: healthcare software, embedded systems, automation or secure computing. These are teaching annotations, not an employer guarantee of eligibility.

A path to explore after further study; graduate-level requirements are retained.

DS: Named discipline appears in source; consult qualifications for eligibility

CS: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Machine learning, Statistics, Cloud, Verification/validation, Genomics/bioinformatics, Communication/teamwork

Eligibility excerpts not shown above

Compensation (source excerpts): Not stated in captured posting text.

Course sheet tags

Majors (teaching tags): DS; CS; Career stage (teaching tag): Research / graduate training path; Geography: DFW; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full-time.

Source record

Original posting https://jobs.utsouthwestern.edu/job/23689439/computational-research-scientist-simmons-cancer-center-dallas-tx/

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Saved raw source: sources/postings/utsw_23689439.json

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